

Cosmos 3: Omnimodal World Models for Physical AI

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Introduction

Motivation: Physical AI requires scalable training environments

- Physical AI = perception + reasoning + action
- Real-world training is not scalable
 - Use simulation-based world models
 - Learn: **understanding** & **generation**

Introduction

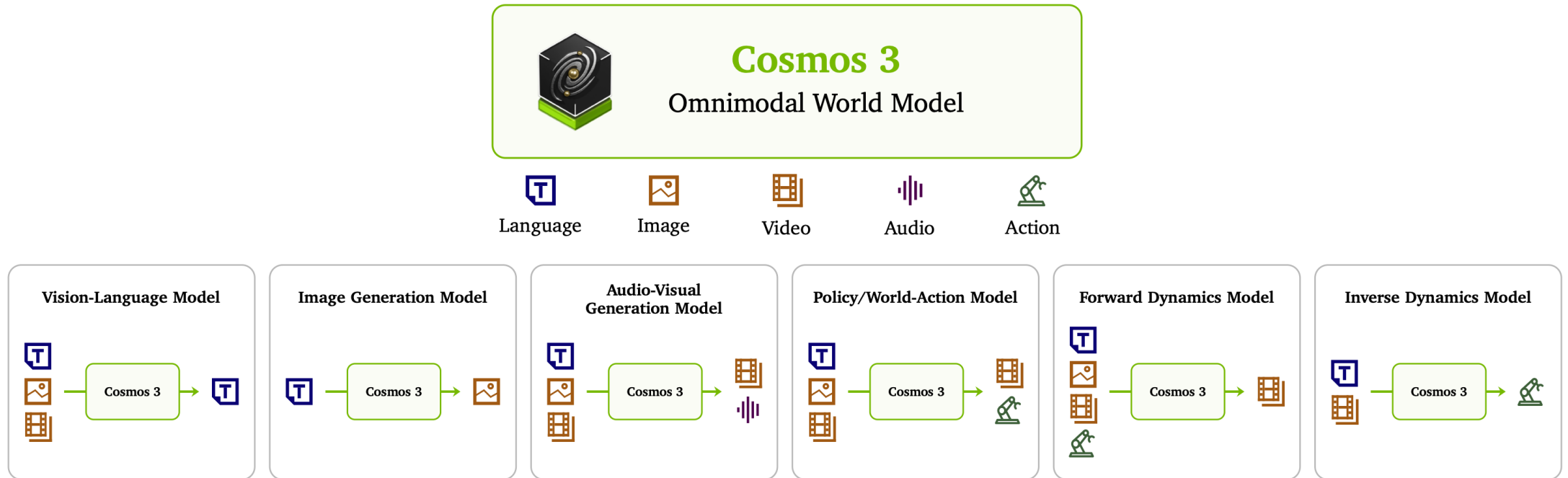
Limitation of prior work: Existing approaches are **fragmented**

- Discriminative models (perception, reasoning)
- Generative Models
- Action Models

Introduction

Limitation of prior work: Existing approaches are **fragmented**

- Cosmos3: **One model** for **perception**, **generation**, and **execution**



Overview of Cosmos3

- Unified multimodal model (language, vision, audio, **action**)
- Modality-specific encoder → shared representation space
- Mixture-of-Transformers (MoT) backbone
- Generation:
 - Language → autoregressive (next-token prediction)
 - Others → diffusion (iterative denoising)

Encoder

Modality-specific encoder → shared representation space

Image and Video

- ViT encoder → **understanding**
 - Pretrained vision-language model
 - Jointly trained with backbone
- VAE encoder → **generation**
 - Spatial + temporal compression
 - Frozen during training

Audio

- Audio VAE encoder

Encoder

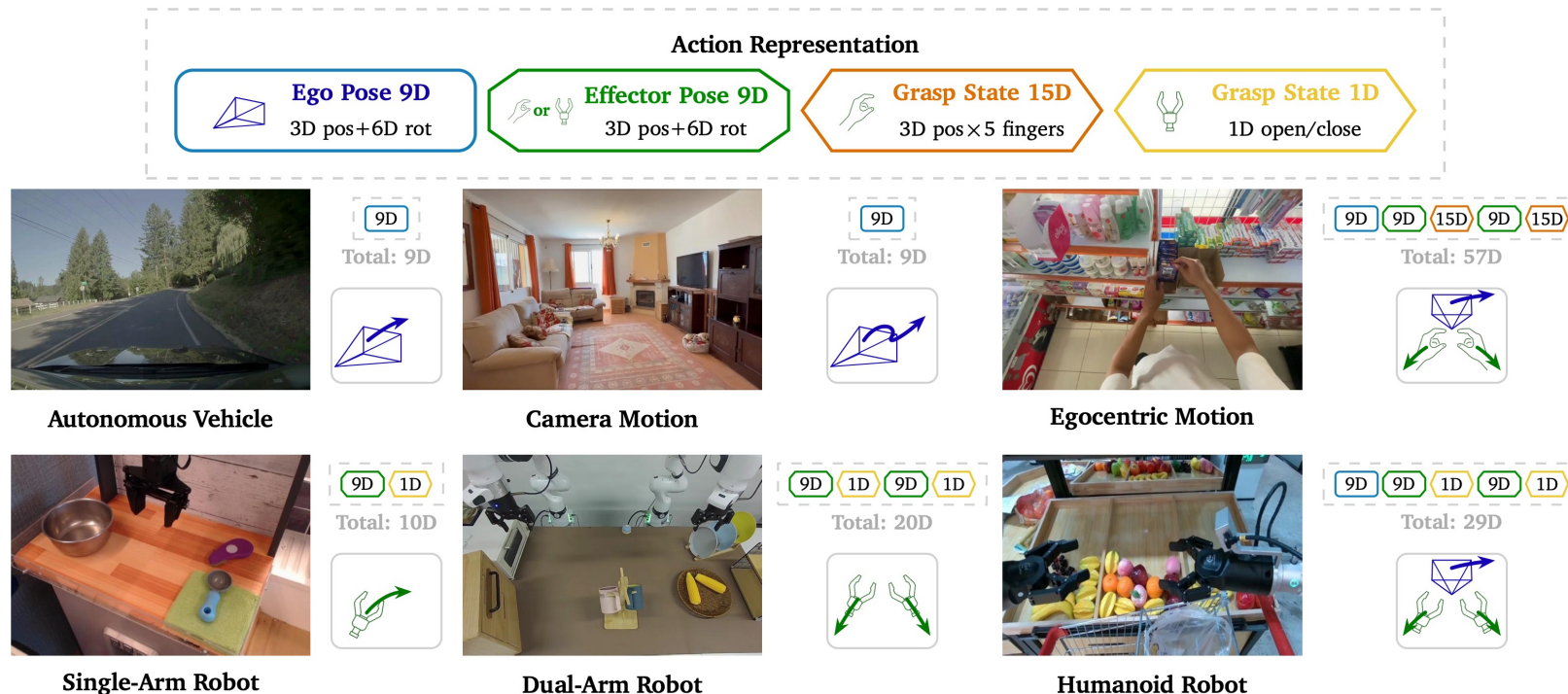
Action: Action Representation

- Heterogenous control signals → cannot be directly modeled together

- **Action** = transition between world states

$$\Delta T_t = T_{t-1}^{-1} T_t \quad (T: \text{poses}, t: \text{time})$$

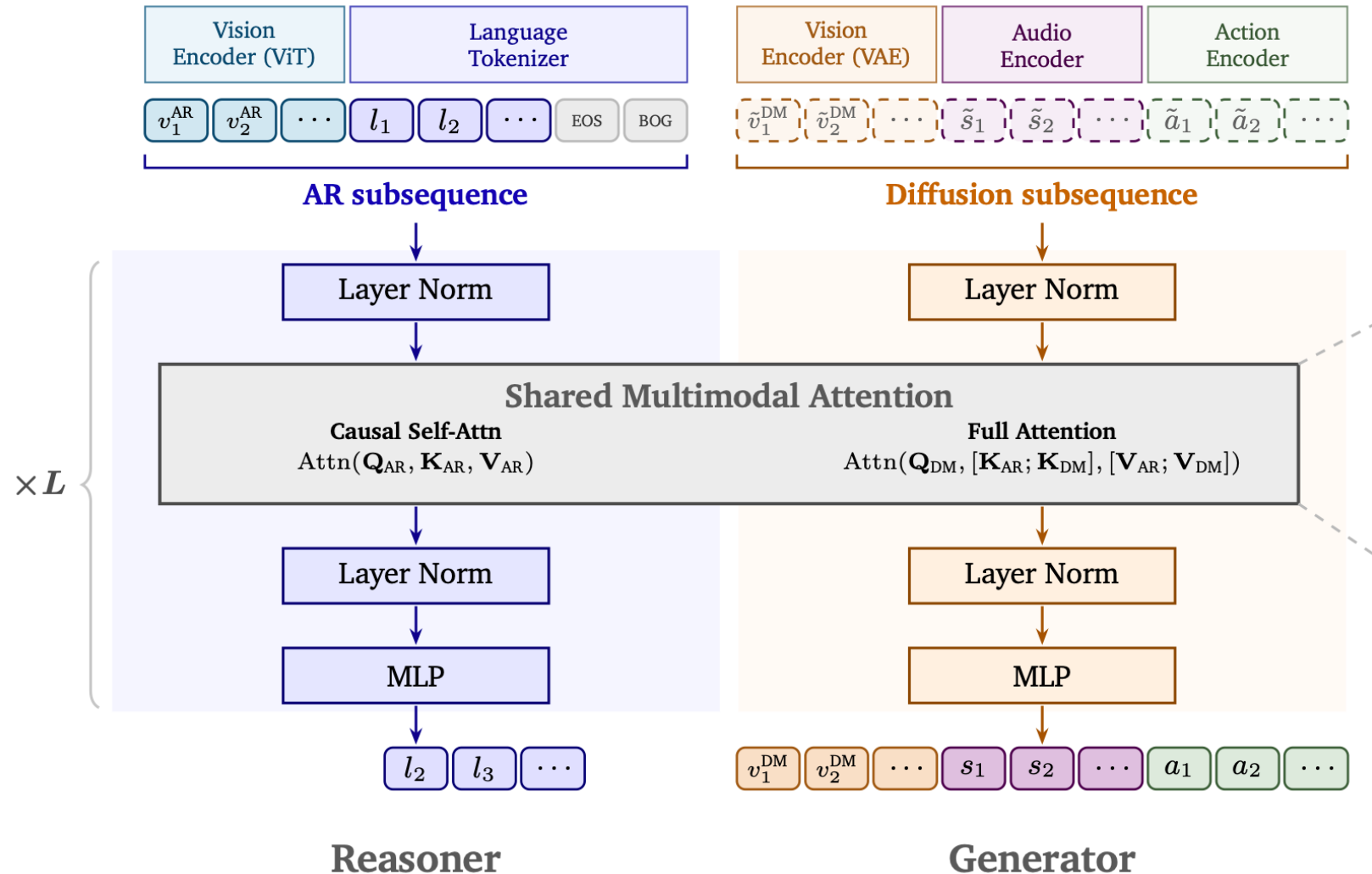
- Components: 1) Ego poses 2) Effector poses 3) Grasp poses



Mixture of Transformer (MoT)

(1) Dual-Tower Layer Structure

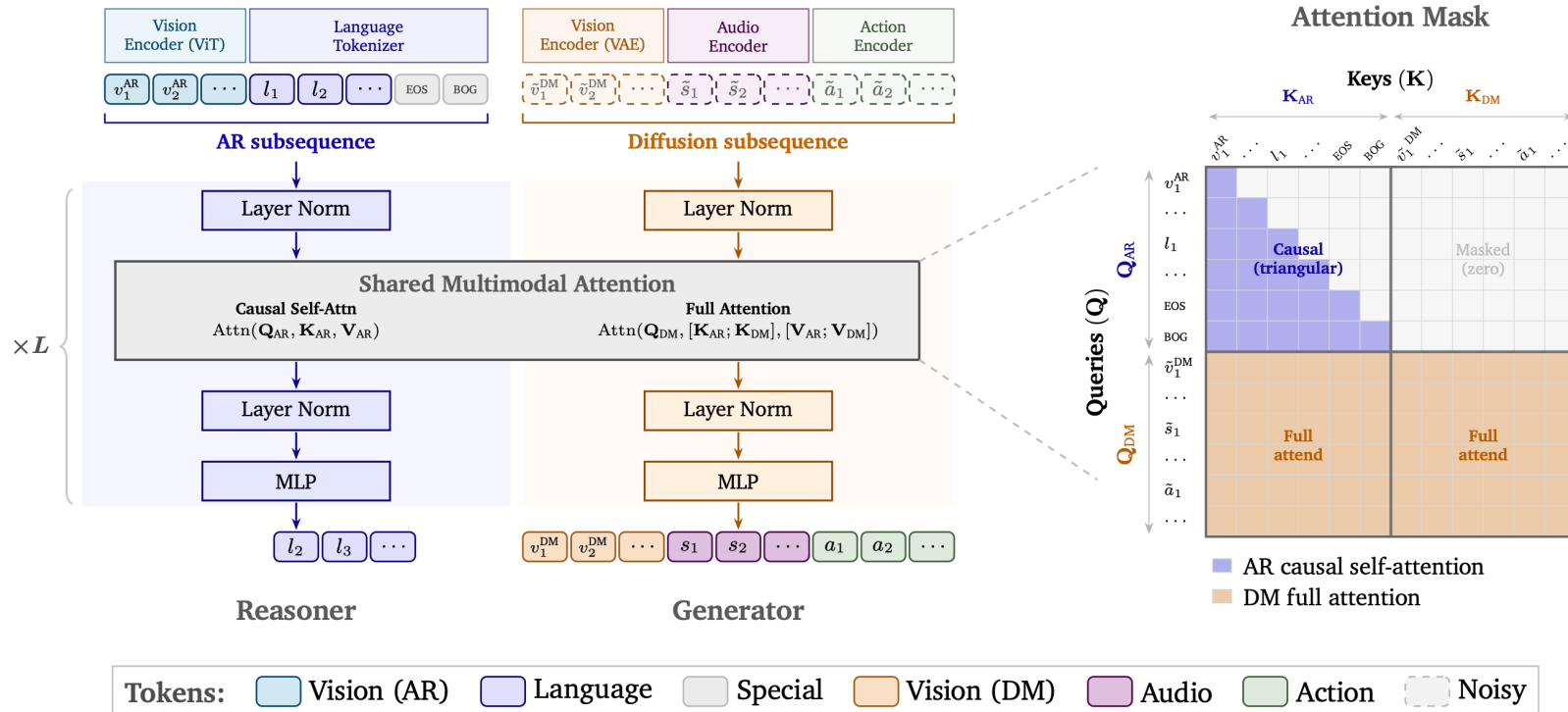
- Shared Self-Attention
- Separate LayerNorm
- Separate MLP



Mixture of Transformer (MoT)

(2) Dual-Stream Joint Attention

- AR (Reasoner) – causal self-attention
- DM (Generator) – Full Attention



Mixture of Transformer (MoT)

(2) Dual-Stream Joint Attention

- **AR (Reasoner)**

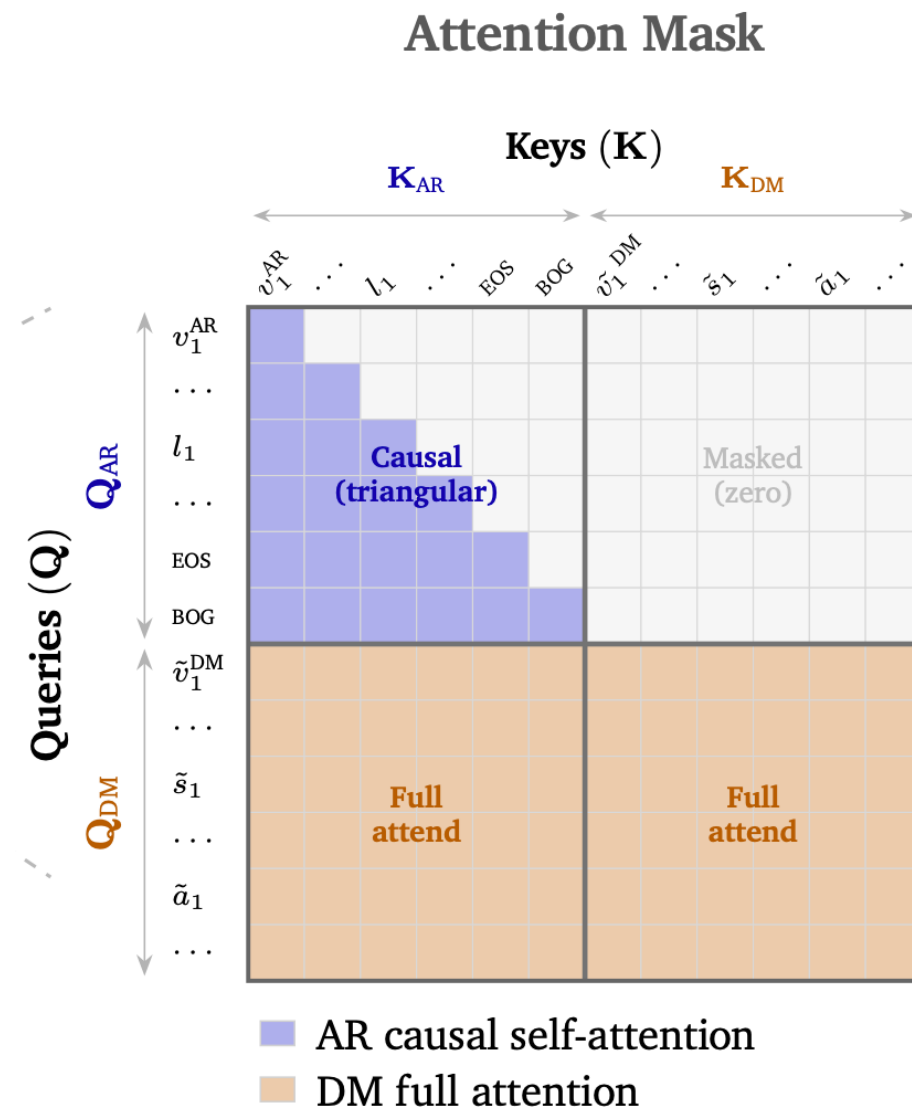
- Output: next-token prediction

$$\mathbf{O}_{\text{AR}} = \text{Attn}_{\text{causal}}(\mathbf{Q}_{\text{AR}}, \mathbf{K}_{\text{AR}}, \mathbf{V}_{\text{AR}})$$

- **DM (Generator)**

- Output: de-noised tokens for generator

$$\mathbf{O}_{\text{DM}} = \text{Attn}_{\text{full}}(\mathbf{Q}_{\text{DM}}, [\mathbf{K}_{\text{AR}}; \mathbf{K}_{\text{DM}}], [\mathbf{V}_{\text{AR}}; \mathbf{V}_{\text{DM}}])$$



Multimodal Position Embedding

Position Embedding: injects temporal, spatial structure

Challenge:

- Different modalities have different temporal resolutions
- Video (FPS), Audio (Sampling Rate), Action (Control frequency)

Solution: 3D MRoPE + **absolute temporal indexing**

- Align heterogenous modalities on the same physical temporal axis

Data Overview

Reasoner → understanding

- Paired Vision-Language Data (Image-Text / Video-Text)
- QA, Grounding, Temporal Reasoning, Action Understanding

Generator → generation

- Generate and simulation the world
 - Image / Video / Audio / Action data
 - Reconstruction-based learning (No explicit annotation required)
-
- Same Transformer Architecture
 - Different Training data by objective

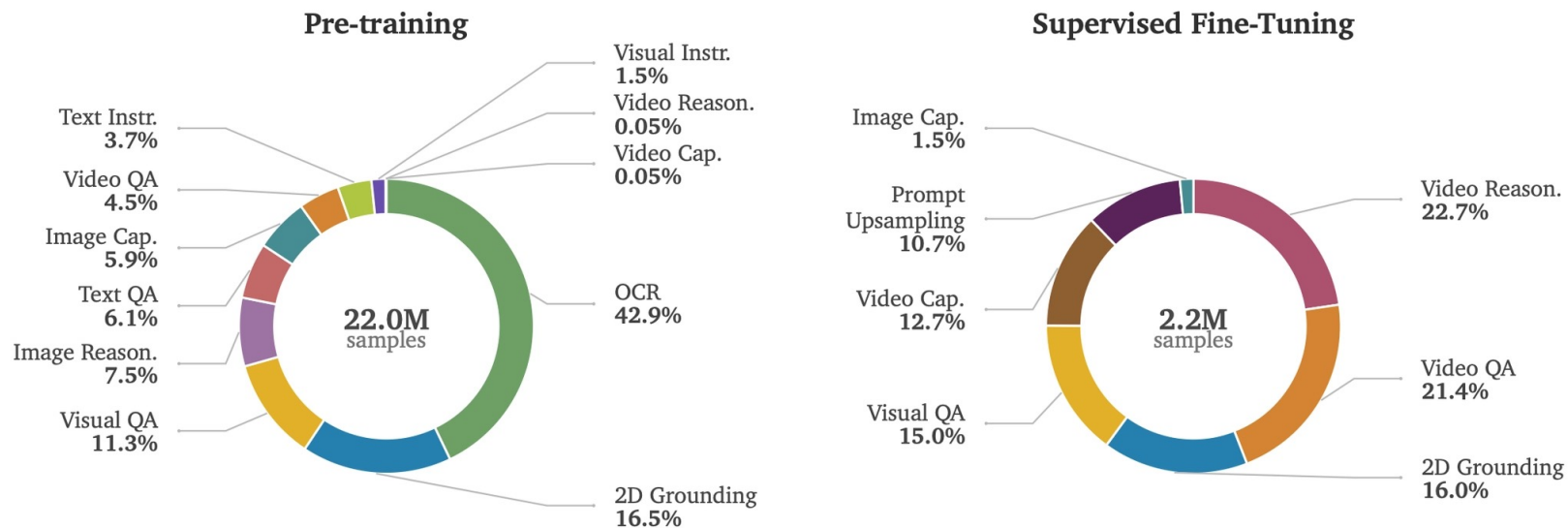
Data - Reasoner

Pre-Training

- Large-scale Vision-Language data (Nemotron Nano2)
- Learn general visual understanding

Supervised Fine-Tuning (SFT)

- Physical AI domain data + Synthetic data
- Robotics / Autonomous Driving / Smart Infrastructure
- Learn Grounding / Spatial Understanding / Temporal Understanding



Data - Generator

Pre-Training

- Learn general world generation capability
- Large-scale Multimodal Data

Mid Training

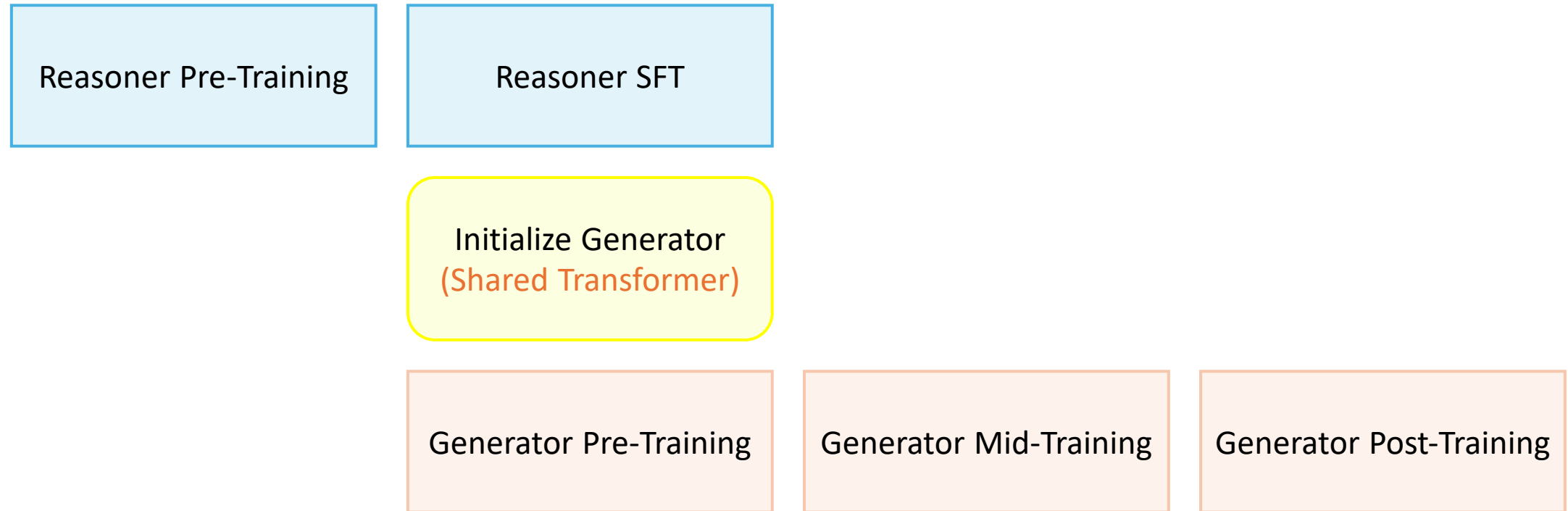
- High-quality curated Data
- Physical AI synthetic Data
- Transfer Data, Action Data

Post Training

- Domain-specific expert datasets → fine-tuning
- Create task-specific expert models (T2I / I2V / Policy expert)

Training

Overall Training Pipeline



Training

Generator Training Objective

- Goal: Unified generation across modalities (Image/Video/Audio/Action)
- Rectified Flow Matching:

$$x_{\sigma} = \sigma\epsilon + (1 - \sigma)x_0 \quad (x_0: \text{clean data}, \epsilon: \text{random noise})$$
$$v^* = \epsilon - x_0 \quad (v: \text{velocity})$$

- Add noise to target sample
- **Predict velocity** from noisy sample to clean sample
- Same objective across all modalities

Conclusion

- Cosmos3 unifies multimodal **understanding** and **generation** across:
 - Language
 - Image
 - Video
 - Audio
 - Action
 - Within a **single architecture**
- Impact
 - Bridge between simulation and real-world AI
 - Enables scalable synthetic data generation
 - Unified foundation model for Physical AI
 - Supports closed-loop embodied learning